ProyectoAED

Importación de los DataFrames

```
library(readr)
library(stringr)
library(dplyr)
library(tidyr)
library(magrittr)
library(lubridate)
library(ggplot2)
library(GGally)
```

```
# Cargamos el dataframe
poblacion <- read_tsv('data/poblacion_paises.tsv')

# Limpiamos los nombres de la variable country para obtener los nombres de los paises
colnames(poblacion)[1] <- 'country'
poblacion %<>% mutate(country = sub("^.*,", "", country)) %>% mutate(across(-1, ~
↪  as.numeric(gsub('[^0-9.]', '', .x))))

# Calculamos la media poblacional de cada pais desde el 2000 hasta hoy
poblacion %<>% transmute(country, population = rowMeans(select(., -country), na.rm =
↪  TRUE))

# Ordenamos los paises de mayor a menor poblacion
# Eliminamos algunos que no pertenecen politicamente a Europa
poblacion %<>% arrange(desc(population)) %>% filter(!(country %in% c('RU', 'UA',
↪  'DE_TOT', 'TR', 'FX', 'BY', 'AZ', 'UK')))

# Nos quedamos con los nombres de los 10 primeros
mas.poblacion10 <- poblacion$country[1:10]

# Nos quedamos con los nombres de los 5 primeros
mas.poblacion5 <- poblacion$country[1:5]
```

```
# Cargamos el dataframe
indep.resid <- read_tsv('data/edad_independencia_residencial.tsv')

# La primera columna viene con los valores de sex y country, por lo que hay que
↪  separarlos
colnames(indep.resid)[1] <- 'sex.country'
indep.resid %<>% mutate(sex.country = sub("^.*,([^,]*,[^,]*)$", "\\1", sex.country)) %>%
↪  separate(sex.country, into = c('sex', 'country'), sep = ',') %>% mutate(sex =
↪  factor(sex))
```

```r
# Aqui pasamos los valores de edad de emancipacion en funcion de año a formato tidy
# Filtramos para quedarnos con sex = 'T', el total, ya que no nos interesa la diferenciacion por sexos
# Nos quedamos con los 10 paises mas poblados
indep.resid %<>% mutate(across(c(-1, -2), ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
  pivot_longer(cols = c(-1, -2), names_to = 'year', values_to = 'age.indep') %>%
  mutate(year = as.numeric(year)) %>% filter(sex == 'T' & country %in% mas.poblacion10)
  %>% select(-sex) %>% mutate(country = factor(country, levels = mas.poblacion10,
  labels = mas.poblacion10))
```

```r
precio.vivienda <- read_tsv('data/precio_vivienda.tsv')

colnames(precio.vivienda)[1] <- 'country'
precio.vivienda %<>% mutate(country = sub(".*,", "", country))

precio.vivienda %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
  pivot_longer(cols = -1, names_to = 'year.quadri', values_to = 'rel.price2015.house')

# Como el indice viene por cuatrimestres, nos quedamos con 1 solo valor, el correspondiente al cuatrimestre 1
precio.vivienda %<>% separate(year.quadri, into = c('year', 'quadri'), sep = '-') %>%
  mutate(quadri = factor(quadri, ordered=TRUE)) %>% mutate(year = as.numeric(year)) %>%
  filter(country %in% mas.poblacion10 & quadri == 'Q1') %>% select(-quadri) %>%
  mutate(country = factor(country, levels = mas.poblacion10, labels = mas.poblacion10))
```

```r
precio.alquiler <- read_tsv('data/precio_alquiler.tsv')

colnames(precio.alquiler)[1] <- 'country'
precio.alquiler %<>% mutate(country = sub(".*,", "", country))

precio.alquiler %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>%
  pivot_longer(cols = -1, names_to = 'year', values_to = 'rel.price2015.rent') %>%
  mutate(year = as.numeric(year)) %>% filter(country %in% mas.poblacion10) %>%
  mutate(country = factor(country, levels = mas.poblacion10, labels = mas.poblacion10))
```

```r
salarios2 <- read_tsv('data/salarios2.tsv')

# Este dataframe ha sido obtenido de otra fuente
# Por eso los nombres de los paises vienen en otro formato y hay que hacer el filtrado manualmente otra vez
salarios2 %<>% transmute(country = ref_area.label, year = time, earnings = 0.9*obs_value,
  source = factor(source.label))
names.country <- c('Germany', 'France', 'Italy', 'Spain', 'Poland', 'Romania',
  'Netherlands', 'Belgium', 'Greece', 'Czechia')
names.country %in% salarios2$country
salarios2 %<>% filter(country %in% names.country) %>% mutate(country = factor(country,
  levels = names.country, labels = mas.poblacion10))

# En algunos paises se ha cambiado el metodo de cuantificar los salarios, por lo que hay saltos muy grandes de un año a otro
# Identificamos los paises en los que ha pasado esto
major.source <- salarios2 %>% count(country, source) %>% group_by(country) %>%
  arrange(desc(n)) %>% slice(1) %>% mutate(major.source = source) %>% select(c(-source,
  -n))
```

```r
# Para no estropear la tendencia temporal, nos quedamos con los datos correspondientes a
↪  1 solo metodo, del que mas observaciones haya
salarios2 %<>% full_join(major.source) %>% mutate(earnings = ifelse(source ==
↪  major.source, earnings, NA)) %>% select(c(-major.source, -source))
```

```r
ipca <- read_tsv('data/ipca.tsv')

colnames(ipca)[1] <- 'country'
ipca %<>% mutate(country = sub(".*,", "", country))

ipca %<>% mutate(across(-1, ~ as.numeric(gsub('[^0-9.]', '', .x)))) %>% pivot_longer(cols
↪  = -1, names_to = 'year', values_to = 'rel.value2025.ipca') %>% mutate(year =
↪  as.numeric(year)) %>% mutate(country = factor(country, levels = mas.poblacion10,
↪  labels = mas.poblacion10))
```

Transformaciones de variables

```r
# Juntamos todas las variables en un dataframe por paises y años
jovenes.df <- indep.resid %>% full_join(salarios2) %>% left_join(ipca) %>%
↪  full_join(precio.alquiler) %>% full_join(precio.vivienda)
```

```
## Joining with `by = join_by(country, year)`
## Joining with `by = join_by(country, year)`
## Joining with `by = join_by(country, year)`
## Joining with `by = join_by(country, year)`
```

```r
# Obtenemos el primer año en el que comienzan las medidas de salarios y precios
# Tomamos este año como valor de referencia para poder compararlos luego mas facilmente
year.start <- jovenes.df %>% group_by(country) %>% arrange(year) %>% select(c(country,
↪  year, rel.price2015.rent, rel.price2015.house, earnings)) %>%
↪  filter(if_all(everything(), ~ !is.na(.x))) %>% slice(1) %>% ungroup() %>%
↪  transmute(country, year.start = year)

jovenes.df <- full_join(jovenes.df, year.start)
```

```
## Joining with `by = join_by(country)`
```

```r
# Obtenemos el aumento real de los salarios, descontando la inflacion
jovenes.df %<>% group_by(country) %>% mutate(real.earnings =
↪  earnings*rel.value2025.ipca[year==unique(year.start)]/rel.value2025.ipca) %>%
↪  ungroup() %>% select(-rel.value2025.ipca)

# Ahora tomamos, para cada variable, un valor de 100 en el año de referencia, asi las
↪  podremos comparar mejor graficamente
jovenes.df %<>% group_by(country) %>% mutate(earnings = earnings/earnings[year ==
↪  unique(year.start)]*100, real.earnings = real.earnings/real.earnings[year ==
↪  unique(year.start)]*100, price.rent = rel.price2015.rent/rel.price2015.rent[year ==
↪  unique(year.start)]*100, price.house = rel.price2015.house/rel.price2015.house[year
↪  == unique(year.start)]*100) %>% select(c(-year.start, -rel.price2015.rent,
↪  -rel.price2015.house)) %>% ungroup()
```

```r
# Calculamos los salarios reales en proporcion al precio de la vivienda y a los
↪  alquileres, respectivamente
jovenes.df %<>% mutate(earn.prop.house = real.earnings/price.house*100, earn.prop.rent =
↪  real.earnings/price.rent*100)

# Calculamos el aumento porcentual de los salarios, alquileres y precio de la vivienda
↪  para cada año
jovenes.df %<>%
  group_by(country) %>%
  arrange(year, .by_group = TRUE) %>%
  mutate(
    earn.diff = earnings - lag(earnings),
    real.earn.diff = real.earnings - lag(real.earnings),
    house.diff = price.house - lag(price.house),
    rent.diff = price.rent - lag(price.rent),
  ) %>%
  ungroup()
```

Estudio de variables

```r
sumario.jovenes <- data.frame(variable = colnames(jovenes.df))
sumario.jovenes$type <- sapply(jovenes.df, function(x) class(x)[1])
sumario.jovenes$levels <- sapply(jovenes.df, function(x) length(unique(x)))
sumario.jovenes$topLevel <- sapply(jovenes.df, function(x) names(which.max(table(x))))
sumario.jovenes$topCount <- sapply(jovenes.df, function(x) max(table(x)))
sumario.jovenes$topFrac <- sumario.jovenes$topCount / nrow(jovenes.df)
sumario.jovenes$missFreq <- sapply(jovenes.df, function(x) sum(is.na(x)))
sumario.jovenes$missFrac <- sumario.jovenes$missFreq / nrow(jovenes.df)

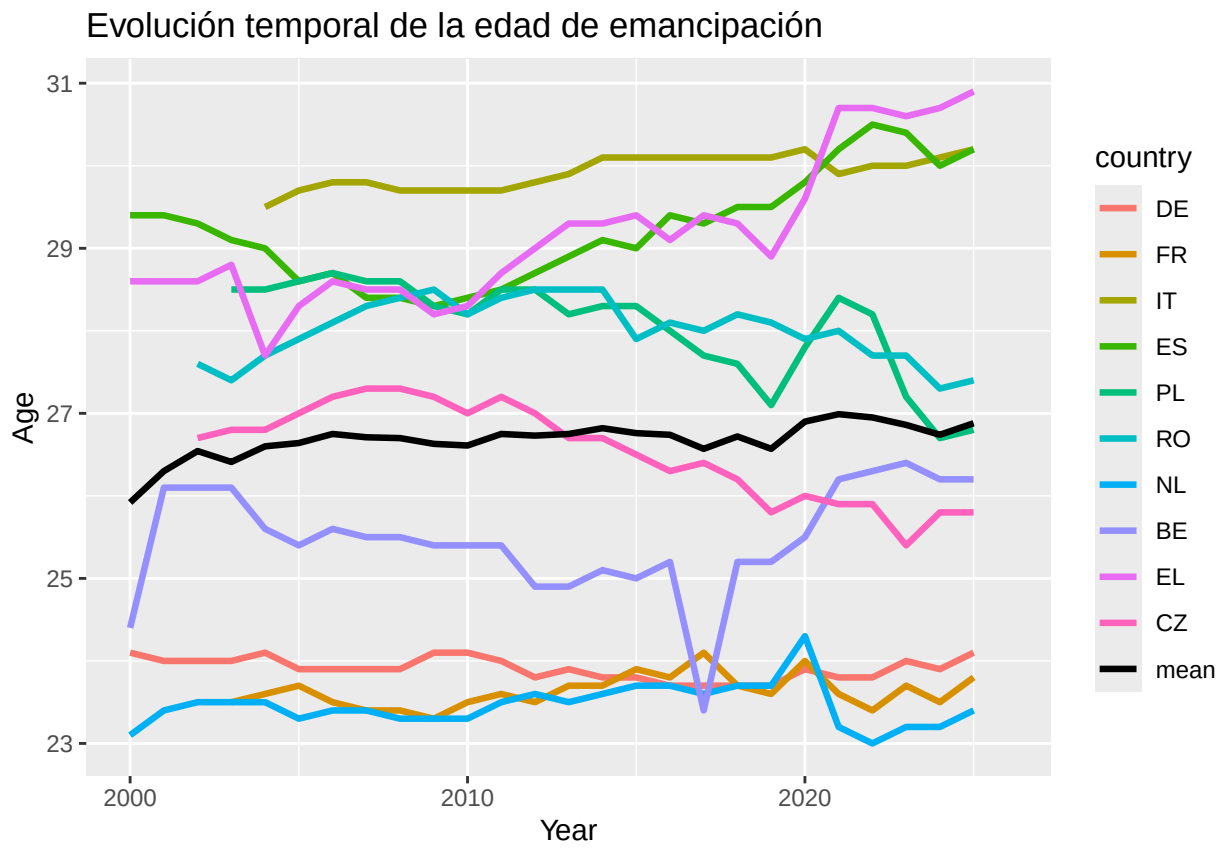
sumario.jovenes
```

```
##           variable    type levels          topLevel topCount     topFrac
## 1          country  factor     10                DE       27 0.100371747
## 2             year numeric     27              2000       10 0.037174721
## 3        age.indep numeric     68              23.7       13 0.048327138
## 4         earnings numeric    168               100        9 0.033457249
## 5    real.earnings numeric    168               100        9 0.033457249
## 6       price.rent numeric    225               100        9 0.033457249
## 7      price.house numeric    172               100        9 0.033457249
## 8  earn.prop.house numeric    155               100        9 0.033457249
## 9   earn.prop.rent numeric    168               100        9 0.033457249
## 10       earn.diff numeric    167 -42.8651318317234        1 0.003717472
## 11  real.earn.diff numeric    167  -36.429192556161        1 0.003717472
## 12      house.diff numeric    169  1.45547945205479        3 0.011152416
## 13       rent.diff numeric    213  1.12612612612612        3 0.011152416
##    missFreq   missFrac
## 1         0 0.00000000
## 2         0 0.00000000
## 3        23 0.08550186
## 4        94 0.34944238
```

```
## 5         94 0.34944238
## 6         36 0.13382900
## 7         89 0.33085502
## 8        107 0.39776952
## 9         94 0.34944238
## 10       103 0.38289963
## 11       103 0.38289963
## 12        98 0.36431227
## 13        45 0.16728625
```

Análisis univariante

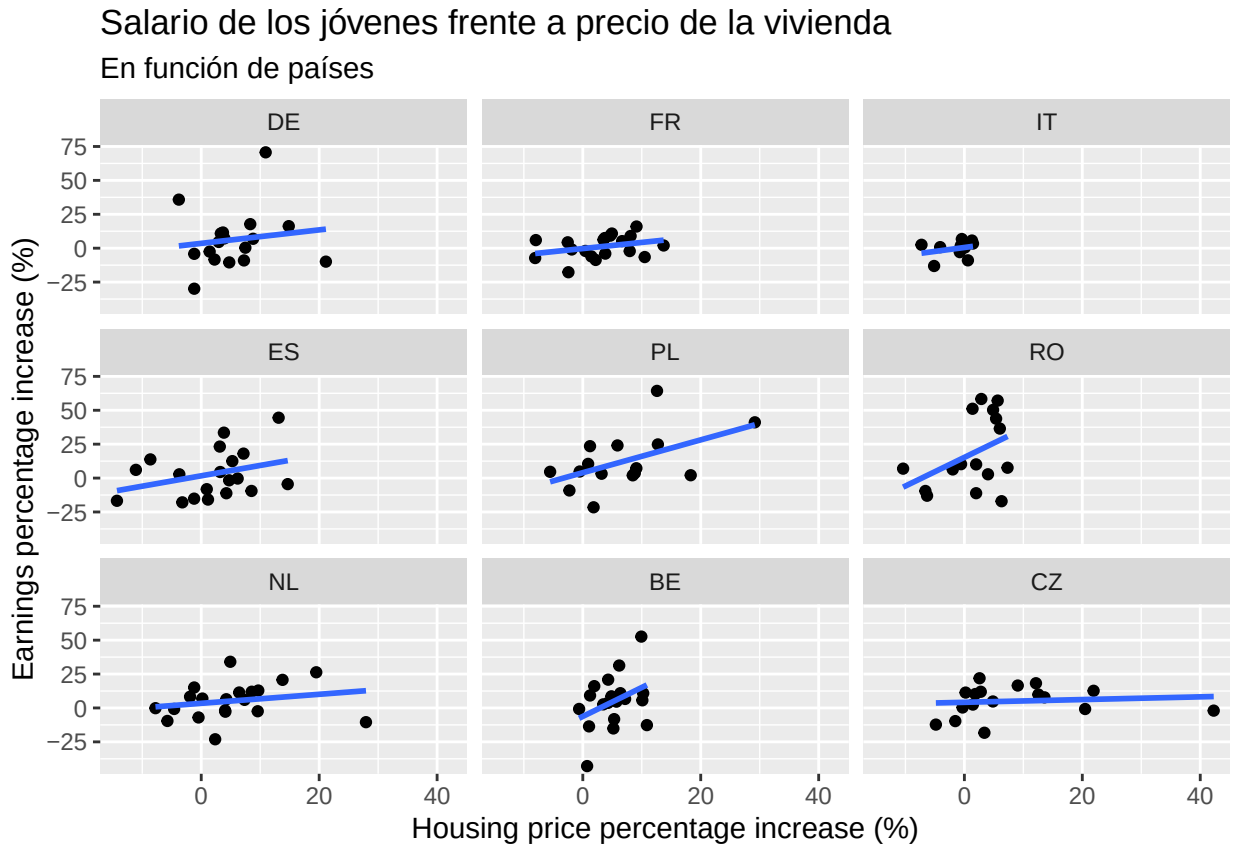
```
age.mean <- jovenes.df %>% group_by(year) %>% summarise(age.mean = mean(age.indep, na.rm
↪  = TRUE), .groups = "drop")
ggplot() + geom_line(data = jovenes.df, mapping = aes(x = year, y = age.indep, color =
↪  country), linewidth = 1.15, na.rm = TRUE) + geom_line(data = age.mean, mapping =
↪  aes(x = year, y = age.mean, color = 'mean'), linewidth = 1.15, na.rm = TRUE) +
  scale_color_manual(values =
   ↪  c(setNames(scales::hue_pal()(length(unique(jovenes.df$country))),
   ↪  unique(jovenes.df$country)), "mean" = "black")) +
  labs(title = 'Evolución temporal de la edad de emancipación') + xlab('Year') +
   ↪  ylab('Age')
```



```
ggsave('results/edad_emancipacion.pdf', width = 14, height = 10, dpi = 300)
```

Análisis bivariante

```
jovenes.df %>% filter(country != 'EL', year >= 2005 & year <= 2025) %>% ggplot(aes(x =
↪  house.diff, y = earn.diff)) + geom_point(na.rm = TRUE) +
  geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nrow = 4)
  ↪  + ylab('Earnings percentage increase (%)') + xlab('Housing price percentage
  ↪  increase (%)') + labs(title = 'Salario de los jóvenes frente a precio de la
  ↪  vivienda', subtitle = 'En función de países')
```

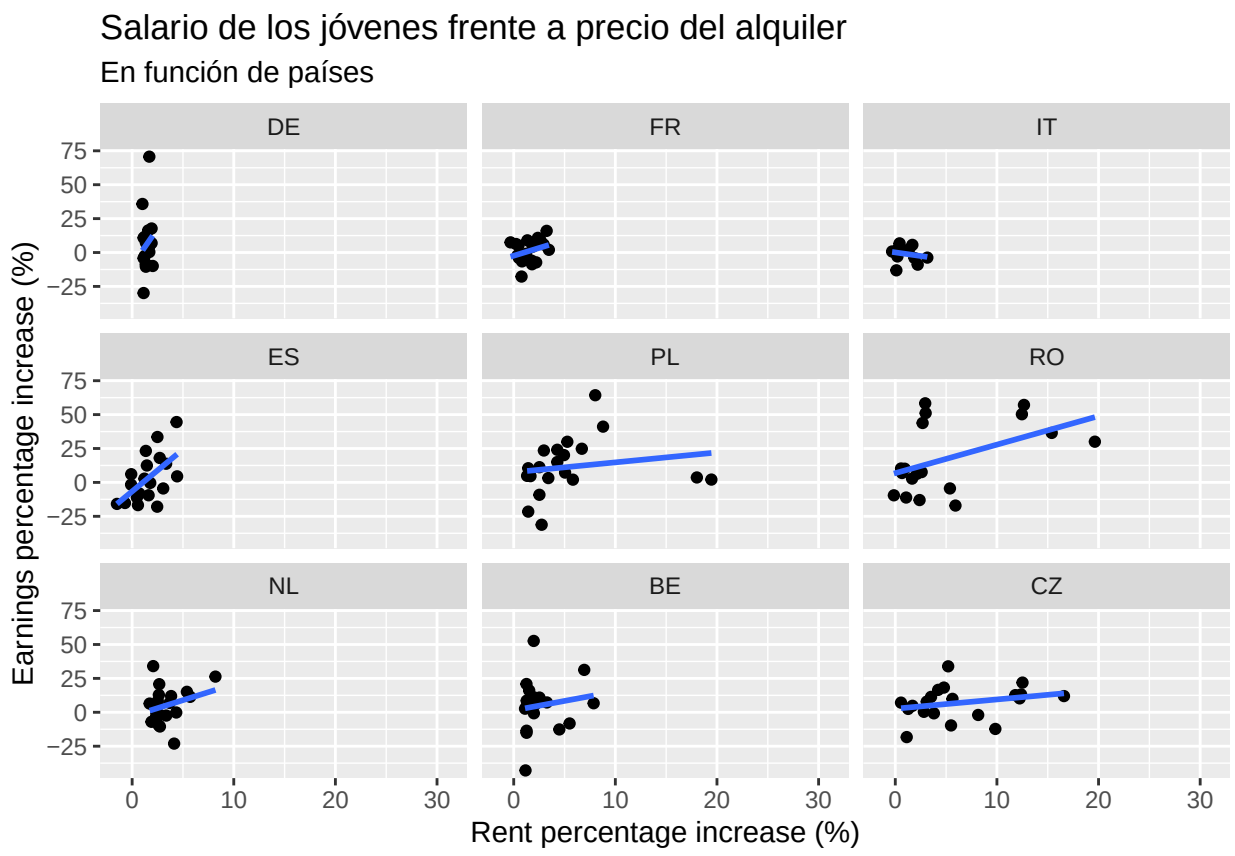



```
jovenes.df %>% filter(country != 'EL', year >= 2005 & year <= 2025) %>% group_by(country)
↪  %>% summarise(cor.pearson = cor(real.earnings, price.house, method = 'pearson', use =
↪  'complete.obs'), cor.spearman = cor(real.earnings, price.house, method = 'spearman',
↪  use = 'complete.obs'))
```

```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE            0.774        0.437
## 2 FR           -0.592       -0.565
## 3 IT            0.852        0.655
```

```
## 4 ES            0.392        0.423
## 5 PL            0.779        0.768
## 6 RO            0.874        0.902
## 7 NL            0.144        0.355
## 8 BE           -0.233       -0.305
## 9 CZ            0.308        0.340
```

```
jovenes.df %>% filter(country != 'EL', year >= 2005 & year <= 2025) %>% ggplot(aes(x =
↪  rent.diff, y = earn.diff)) + geom_point(na.rm = TRUE) +
  geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) + facet_wrap(.~country, nrow = 4)
  ↪  + ylab('Earnings percentage increase (%)') + xlab('Rent percentage increase (%)') +
  ↪  labs(title = 'Salario de los jóvenes frente a precio del alquiler', subtitle = 'En
  ↪  función de países')
```

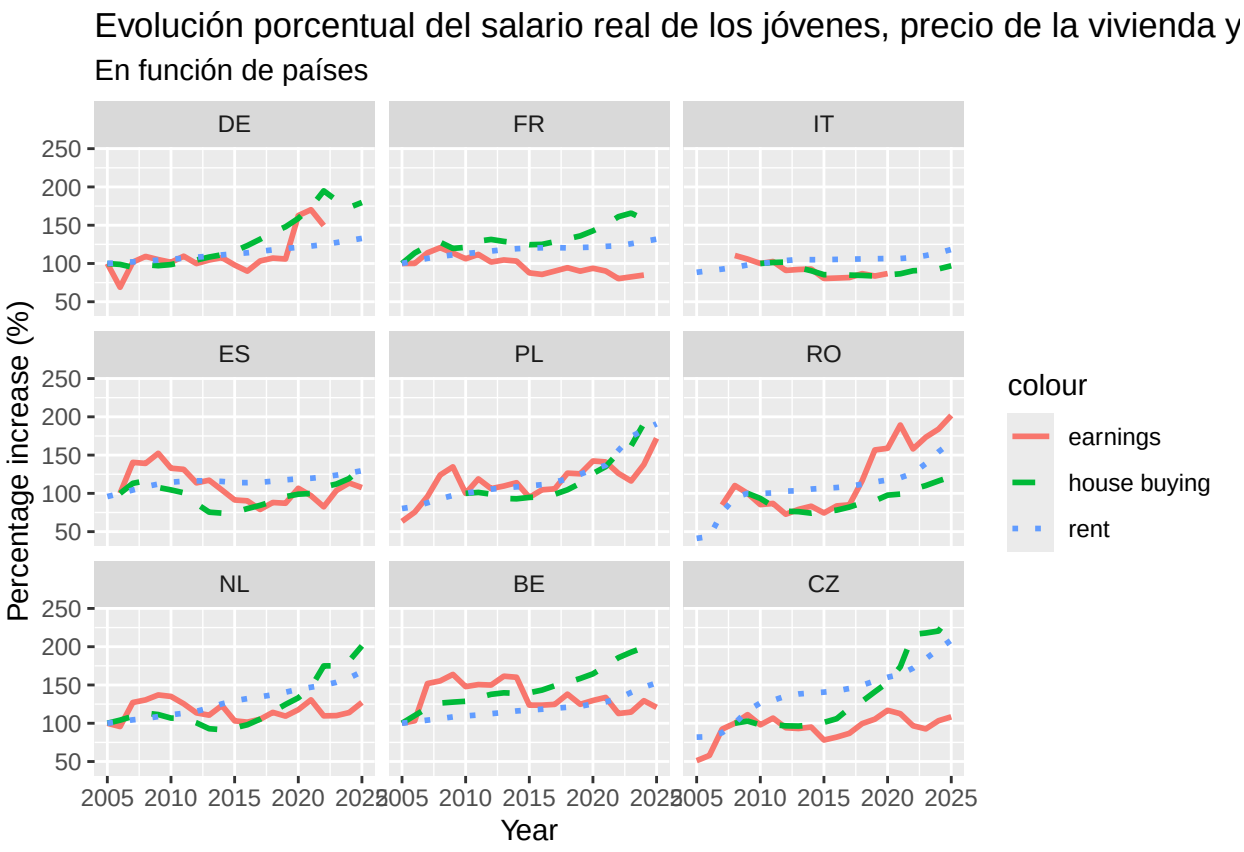



```
jovenes.df %>% filter(country != 'EL', year >= 2005 & year <= 2025) %>% group_by(country)
↪  %>% summarise(cor.pearson = cor(real.earnings, price.rent, method = 'pearson', use =
↪  'complete.obs'), cor.spearman = cor(real.earnings, price.rent, method = 'spearman',
↪  use = 'complete.obs'))
```

```
## # A tibble: 9 x 3
##   country cor.pearson cor.spearman
##   <fct>         <dbl>        <dbl>
## 1 DE            0.700        0.523
## 2 FR           -0.699       -0.809
```

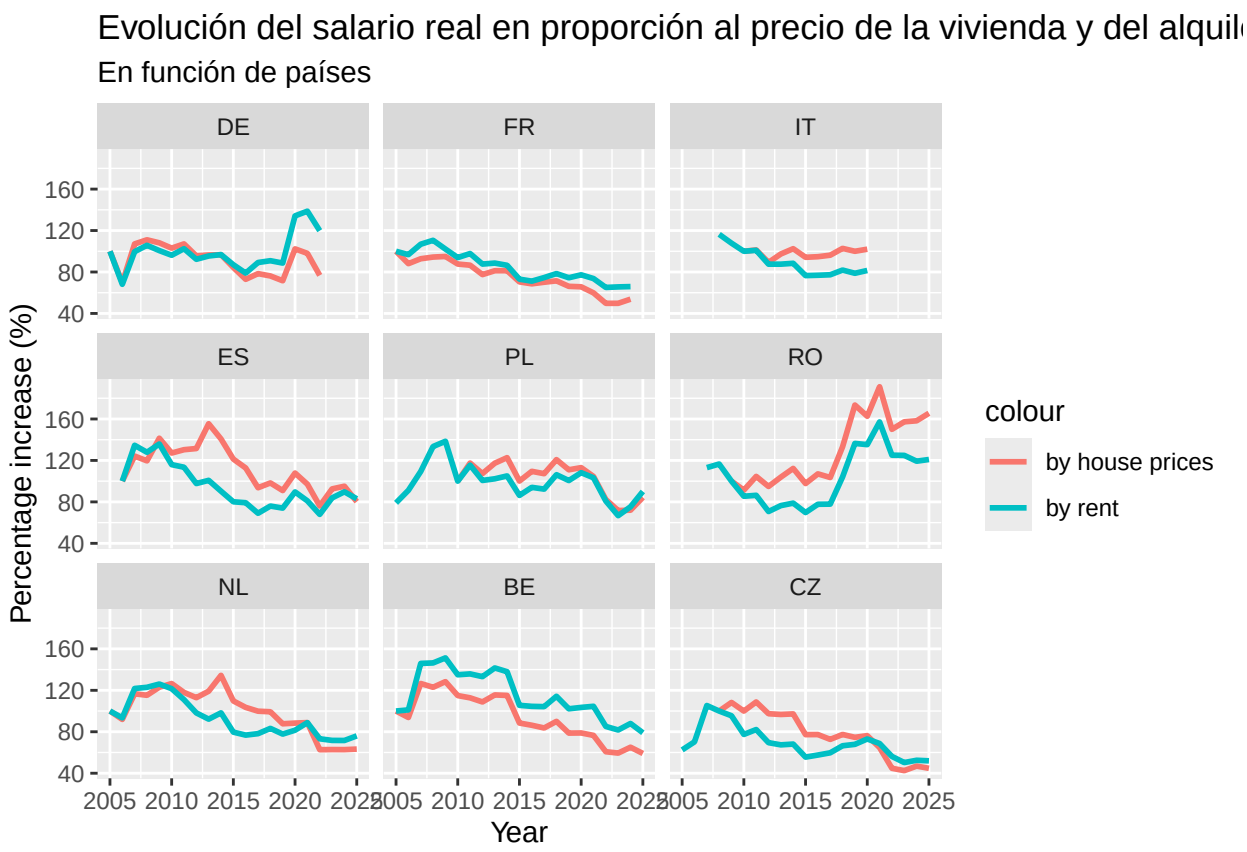
```
## 3 IT          -0.904      -0.753
## 4 ES          -0.279      -0.245
## 5 PL           0.716       0.682
## 6 RO           0.816       0.696
## 7 NL          -0.0583      0.00390
## 8 BE          -0.289      -0.306
## 9 CZ           0.561       0.417
```

```
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = real.earnings, color = 'earnings'), linewidth = 1, na.rm =
  ↪  TRUE) +
  geom_line(aes(x = year, y = price.house, color = 'house buying'), linewidth = 1,
  ↪  linetype = 2, na.rm = TRUE) +
  geom_line(aes(x = year, y = price.rent, color = 'rent'), linewidth = 1, linetype = 3,
  ↪  na.rm = TRUE) + facet_wrap(.~country, nrow = 4) + labs(title = 'Evolución
  ↪  porcentual del salario real de los jóvenes, precio de la vivienda y precio del
  ↪  alquiler', subtitle = 'En función de países') + xlab('Year') + ylab('Percentage
  ↪  increase (%)')
```




```
ggsave('results/viviendas_salarios.pdf', width = 14, height = 10, dpi = 300)
```

```
ggplot(data = filter(jovenes.df, country != 'EL', year >= 2005 & year <= 2025)) +
  geom_line(aes(x = year, y = earn.prop.house, color = 'by house prices'), linewidth = 1,
  ↪  na.rm = TRUE) +
  geom_line(aes(x = year, y = earn.prop.rent, color = 'by rent'), linewidth = 1, na.rm =
  ↪  TRUE) + facet_wrap(.~country, nrow = 4) + labs(title = 'Evolución del salario real
  ↪  en proporción al precio de la vivienda y del alquiler', subtitle = 'En función de
  ↪  países') + xlab('Year') + ylab('Percentage increase (%)')
```




```
ggsave('results/proporcion_salario.pdf', width = 14, height = 10, dpi = 300)
```